Integrated Multiresolution Analysis and Academic Performance System

A sentiment index with an explicit statistical check of its efficiency –and of its limits–, via embeddings and the wavelet transform

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Abstract

We present a system that combines the quantitative signal of a course (grades, midterms, risk status) with the qualitative signal of an open-ended feedback survey, producing for the latter a per-student sentiment index $\hat{\theta} \in (-1, 1)$: each response is turned into a scalar signal indexed by token position (projecting each word's contextual embedding onto the sentence embedding), which is decomposed with the discrete wavelet transform (DWT) into a global-trend component and a local-fluctuation component. We accompany $\hat{\theta}$ with an explicit statistical check –Fisher information and the Cramér–Rao bound under Gaussian and Laplacian models– and show, with simulations, exactly what that check certifies and what it does not. We do not report runs on real course data: this is a methods article, accompanied by illustrative simulations explicitly labeled as such, not a case study.

Keywords: academic analytics, early warning, sentiment analysis, wavelet transform, Fisher information, Cramér–Rao bound.

Code metadata	**Value**
Current code version	v1.0
Permanent link to code/repository used for this code version	github.com/Gabriel44svg/...-SIAMDA
Legal code license	MIT
Code versioning system used	git
Software code languages, tools, and services used	Python 3.11; Streamlit; sentence-transformers; PyWavelets; SciPy; pandas; Plotly; pytest
Compilation requirements, operating environments and dependencies	Python $\geq$ 3.10; see `requirements.txt` (runtime) and `requirements-dev.txt` (tests)
Link to developer documentation/manual	`README.md` in the repository
Support email for questions	`jcgalindo@up.edu.mx`

Table 1. Code metadata (standard software-paper format).

1. Motivation and significance

A course generates two kinds of signal about a student's performance: a *quantitative* one (midterm grades, homework, presentations) and a *qualitative* one (what the student writes in an open-ended feedback survey). The quantitative signal is the one normally used to detect academic risk; the qualitative one tends to be read impressionistically, with no metric. This work proposes a metric for the second signal and asks whether it adds information about academic risk beyond the first, via its empirical correlation with the final grade.

The construction of $\hat{\theta}$ combines two ingredients that are well established separately: contextual sentence *embeddings* obtained with Siamese-BERT-style networks [8], and the discrete wavelet transform as a multiresolution signal-analysis tool [7]. What the system adds is an explicit statistical validation layer for the group's aggregate sentiment estimator, grounded in the classical theory of estimator efficiency [1, 2, 3], complemented by a normality test [4] and a resampling variance estimate [5]. The contribution of this article is not any of those ingredients on its own, but documenting precisely what their combination certifies –and, above all, what it does *not* certify–, something the first version of our own implementation stated imprecisely (Section 2.2.3).

2. Software description

2.1. Software architecture

The system is organized into four modules:

1. **Ingestion and normalization** (`datos.py`): reading grade files (CSV/Excel) with automatic column detection (student ID, name, midterms `E1,E2,...`, weighted exam score, homework, presentations, validation), handling of mixed text in non-numeric cells (*"retake the final"*, *"on academic reinstatement"*, *"withdrew"*, etc.), and computation of a binary risk flag.
2. **NLP and wavelet extraction** (`nlp_wavelet.py`): sentence- and token-level embeddings (multilingual *sentence-transformers*) over the survey's open-ended responses, and multiresolution decomposition of the resulting positional signal via DWT.
3. **Theoretical validation** (`cramer_rao.py`): Fisher information and the Cramér–Rao bound for the aggregate sentiment index, under Gaussian and Laplacian models, plus a normality test (Shapiro–Wilk) and a bootstrap estimate of the estimator's variance.
4. **Interactive visualization**: a Streamlit interface with pages for data loading, course metrics, sentiment analysis, and theoretical validation, with automatic natural-language interpretations for each metric.

2.2. Software functionalities

2.2.1 Academic risk flag

Let c_i be student i's final grade and $e_i \in \{\text{Normal}, \dots\}$ their validation status (reinstatement, final exam, withdrawal, etc.). The system defines

$$r_i = \mathbf{1}\{c_i < 70\} \ \vee \ \mathbf{1}\{e_i \neq \text{Normal}\},$$

i.e., a student is flagged at risk if their grade falls below the passing threshold (70) or if they have a special status on record, regardless of the numeric grade.

2.2.2 Wavelet-based sentiment index

For each student i, the multilingual *sentence-transformers* model (`paraphrase-multilingual-MiniLM-L12-v2`, dimension $d = 384$) [8] produces, in a single *forward pass*: the sentence embedding $x_i \in \mathbb{R}^d$ (normalized to unit norm) and, for each token $t = 1, \dots, L_i$ of the response, a contextual embedding $u_{i,t} \in \mathbb{R}^d$. From these a scalar signal is built, *indexed by token position in the text* –an axis with genuine temporal adjacency, unlike the embedding's coordinates–, by projecting each token onto the response's overall semantic direction:

$$s_i(t) \ = \ \langle u_{i,t}, x_i \rangle, \qquad t = 1, \dots, L_i.$$

The discrete wavelet transform [7] is applied to this 1D signal, with the Daubechies-4 family and a default decomposition level $J = 3$, bounded in practice by $\min_i$ `pywt.dwt_max_level`$(L_i, \text{wavelet})$ –the shortest response in the batch, not a fixed dimension–:

$$\text{wavedec}(s_i, J) = (c_A^{(i)},\ c_{D,1}^{(i)},\ \dots,\ c_{D,J}^{(i)}),$$

where $c_A^{(i)}$ are the approximation coefficients (the response's global trend, low frequency *in position*) and $c_{D,j}^{(i)}$ the detail coefficients at level j (local word-to-word fluctuation, high frequency). The energies are defined as

$$E_i^{\text{apr}} = \|c_A^{(i)}\|_2^2, \qquad E_i^{\text{det}} = \sum_{j=1}^{J} \|c_{D,j}^{(i)}\|_2^2,$$

and student i's sentiment index is defined as

$$\hat{\theta}_i \ = \ \tanh\left(\frac{E_i^{\text{det}} - E_i^{\text{apr}}}{E_i^{\text{det}} + E_i^{\text{apr}} + \varepsilon}\right), \qquad \varepsilon = 10^{-9},$$

so that $\hat{\theta}_i \in (-1, 1)$: values close to -1 indicate energy concentrated in the approximation (a contextual response, little local fluctuation) and values close to $+1$ indicate energy concentrated in the details (high local variability, interpreted here as localized emotional charge).

Design fix

In an earlier version of this pipeline, the DWT was applied directly to the 384 coordinates of x_i, treating them as if they were a temporal signal. Those coordinates are axes of a learned feature space with no intrinsic ordering or adjacency: permuting them does not change the meaning of the text, but it did change E_i^{apr} and E_i^{det} under that version, so the reading "approximation = stable content, detail = emotion" was not justified by the transform's own mathematics. The formulation above –projecting each token onto the sentence embedding and applying the DWT over *token position*– uses an axis with genuine sequential structure, for which the DWT's split into global trend (approximation) and local fluctuation (detail) does have the usual backing of wavelet theory [7]. Code: `processing/nlp_wavelet.py` (`obtener_embeddings_token`, `señal_por_posicion`, `aplicar_dwt_posicional`). An intermediate version of `obtener_embeddings_token` –after the fix above– did build the positional signal as described here, but called `modelo.encode()` *twice* (once for the sentence embedding, once for the token embeddings), i.e. two *forward passes*, while the docstring and an earlier version of this paragraph claimed both outputs came from "a single forward pass". Both statements were true individually and false together. The current version uses a single call with `output_value=None`, which sentence-transformers documents as the way to obtain all outputs (including `sentence_embedding` and `token_embeddings`) from one *forward pass*, making the claim above true.

2.2.3 Statistical validation: Fisher information and the Cramér–Rao bound

Let $\hat{\theta} = (\hat{\theta}_1, \ldots, \hat{\theta}_N)$ be the group's sample of sentiment indices (the data, not individual estimators of anything), treated as N i.i.d. observations from a distribution with location parameter μ (the group's average sentiment), and $\bar{\theta} = \frac{1}{N}\sum_i \hat{\theta}_i$ the sample-mean estimator of μ. The Cramér–Rao bound [1, 2, 3] lower-bounds the variance of any unbiased estimator of μ:

$$\operatorname{Var}(\bar{\theta}) \;\geq\; \frac{1}{F(\mu)}, \qquad F(\mu) = -\,\mathbb{E}\left[\frac{\partial^2}{\partial \mu^2} \ln p(X;\mu)\right],$$

where $X = (\hat{\theta}_1, \ldots, \hat{\theta}_N)$ denotes the full sample (so $p(X;\mu) = \prod_i p(\hat{\theta}_i;\mu)$ and $F(\mu)$ already incorporates the factor N, as in the explicit formulas below).

Key formulas

Gaussian model. Under $\theta \sim \mathcal{N}(\mu, \sigma^2)$, with $\hat{\mu} = \bar{\theta}$ and $\hat{\sigma}^2$ the unbiased sample variance:

$$F(\mu) = \frac{N}{\hat{\sigma}^2}, \qquad \mathrm{CRB}_\mu = \frac{1}{F(\mu)} = \frac{\hat{\sigma}^2}{N}.$$

Laplacian model. Under $\theta \sim \text{Laplace}(\mu, b)$, with $\hat{\mu} = \text{median}(\hat{\theta})$ and $\hat{b} = \frac{1}{N}\sum_i |\hat{\theta}_i - \hat{\mu}|$:

$$F(\mu) = \frac{N}{\hat{b}^2}, \qquad \text{CRB}_\mu = \frac{\hat{b}^2}{N}.$$

Unlike the Gaussian case, where the sample mean attains the bound *exactly* for every N, the sample median is the MLE of μ under the Laplacian model and only attains the bound *asymptotically* ($N \to \infty$); for finite N its variance can exceed CRB_μ. **Empirical estimator efficiency (EEE).** With $\widehat{\text{Var}}(\bar{\theta})$ estimated by *bootstrap* [5] ($B = 500$ resamples with replacement):

$$\text{EEE} = \min\left(\frac{\text{CRB}_\mu}{\widehat{\text{Var}}(\bar{\theta})},\ 1\right).$$

The system complements this computation with a Shapiro–Wilk normality test [4] on $\hat{\theta}$: if the resulting p-value is below 0.05, the Laplacian model (more robust to heavy tails) is recommended over the Gaussian one, using the sample median –the optimal estimator under that model– instead of the mean.

Remark 2.1. The Cramér–Rao bound certifies something precise and limited: how close the *sample mean* of $\hat{\theta}$ is to being an optimal estimator of itself under an assumed parametric model. It does not certify that $\hat{\theta}$ correctly measures "real sentiment", nor does it substitute for external validation (e.g. against human annotation) of the index.

Gap: what this check does NOT certify

Under the Gaussian model with unknown σ^2 estimated by $\hat{\sigma}^2$, the sample mean is the minimum-variance unbiased estimator of μ *for every* N (Lehmann–Scheffé), with $\text{Var}(\bar{\theta}) = \sigma^2/N$ exactly. That is, under that model the efficiency is 1 by construction, not by merit of the NLP–wavelet pipeline: no alternative estimator of μ could ever beat the sample mean. The comparison the system makes –$\text{CRB}_\mu = \hat{\sigma}^2/N$ against $\widehat{\text{Var}}(\bar{\theta})$ estimated by *bootstrap on the same sample*– is, in the large-N limit, a comparison between two ways of estimating the *same* quantity (σ^2/N) from the *same* data: the plug-in normal formula with $\hat{\sigma}^2$, and resampling. That the two agree (EEE ≈ 1) is therefore, to a large extent, a numerical consistency check between bootstrap and the asymptotic formula for the variance of the mean –something that happens for almost any reasonably large sample, regardless of how well the embedding-and-wavelet pipeline captures actual sentiment–, and is *not* evidence that the NLP-wavelet pipeline extracts the maximum information available in the text. Verifying that would require comparing $\hat{\theta}$ against an alternative feature-extraction pipeline (or against human annotation) on the same text, not merely checking that the sample mean of $\hat{\theta}$ is a good estimator of itself. Section 3 illustrates this numerically: in the simulation, EEE = 100% even though $\hat{\theta}$ *rejects* normality (Shapiro–Wilk $p < 10^{-3}$). The interface's interpretive text (`interpretar_crb` in `interpretacion.py`, and the "Theoretical framework" in `validacion.py`) used to overclaim exactly this and has since been corrected to reflect the check's actual scope.

2.2.4 Correlation with academic performance

As the final step of the analysis, the system computes the Pearson correlation coefficient r [6] between $\hat{\theta}_i$ and each student's grade c_i, together with its associated p-value and sample size N. A significant negative correlation ($r < 0$, $p < 0.05$) is interpreted as evidence that higher emotional charge in the responses is associated with lower performance; a significant positive correlation is interpreted as possible extrinsic motivation. The system classifies the strength of the association using thresholds ($|r| \geq 0.6$ strong, $|r| \geq 0.3$ moderate, $|r| < 0.3$ weak) and explicitly reports when the correlation is not significant, without forcing a risk narrative where there is no statistical evidence.

3. Illustrative example

These figures are simulation, not course data

No real student data was used to generate this section. The signals, embeddings, and grades are synthetic, built to exercise the real code in `processing/nlp_wavelet.py` and `processing/cramer_rao.py` –the same functions the application uses– on controlled inputs, for the sole purpose of showing how to read the plots the system produces and of numerically exhibiting the point made in Section 2.2.3. The correlation in Figure 4 is imposed by construction in the simulation, not observed. Generation code: `generar_figuras_articulo_en.py`.

We simulate $N = 60$ responses of variable length (L_i a uniform integer in $\{12, \ldots, 54\}$ tokens), each with an "emotional intensity" $e_i \sim \text{Gamma}(2,\ 0.5)$ that controls the amplitude of local bursts inserted into a random-walk trend (Figure 1), and a simulated grade $c_i = \text{clip}(92 - 9\, e_i + \eta_i,\ 40,\ 100)$ with $\eta_i \sim \mathcal{N}(0, 6^2)$, to illustrate how to read a negative correlation.

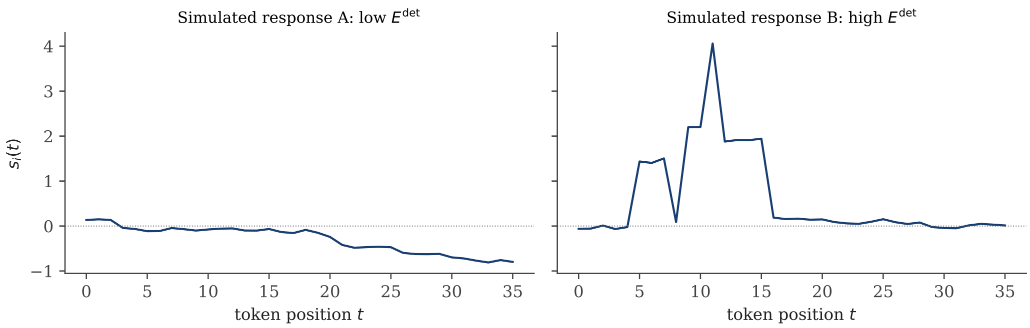

Figure 1. Two simulated positional signals $s_i(t)$: one with low imposed emotional intensity (left) and one with marked local bursts (right), producing low and high E^{det} respectively.

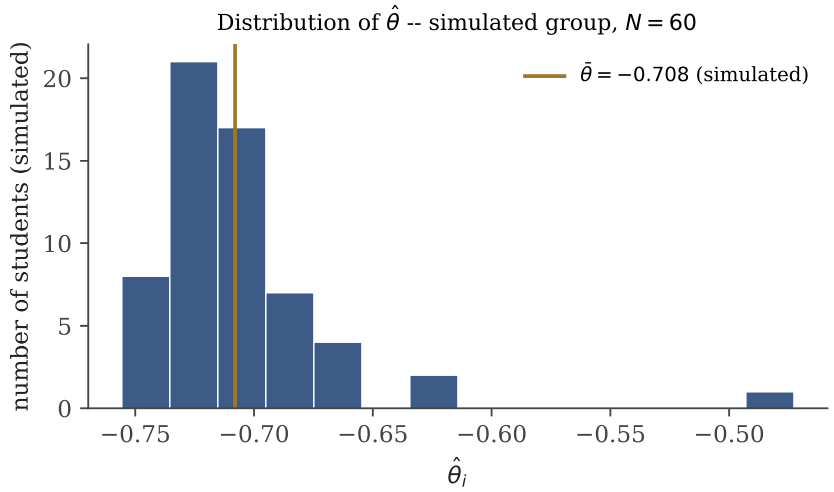

Figure 2. Distribution of $\hat{\theta}$ over the simulated group ($N = 60$). The visible asymmetry is consistent with the rejection of normality reported in the text (Shapiro–Wilk $p < 10^{-3}$).

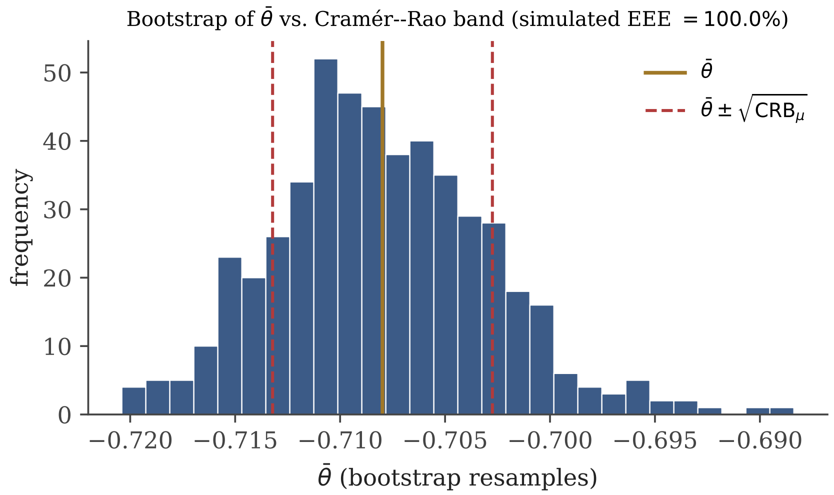

Figure 3. Bootstrap distribution of $\bar{\theta}$ ($B = 500$) against the $\bar{\theta} \pm \sqrt{\mathrm{CRB}_\mu}$ band. In this simulation EEE $= 100\%$ *even though* $\hat{\theta}$ rejects normality –the central point of Section 2.2.3: this figure does not measure whether the pipeline extracts information from the text.

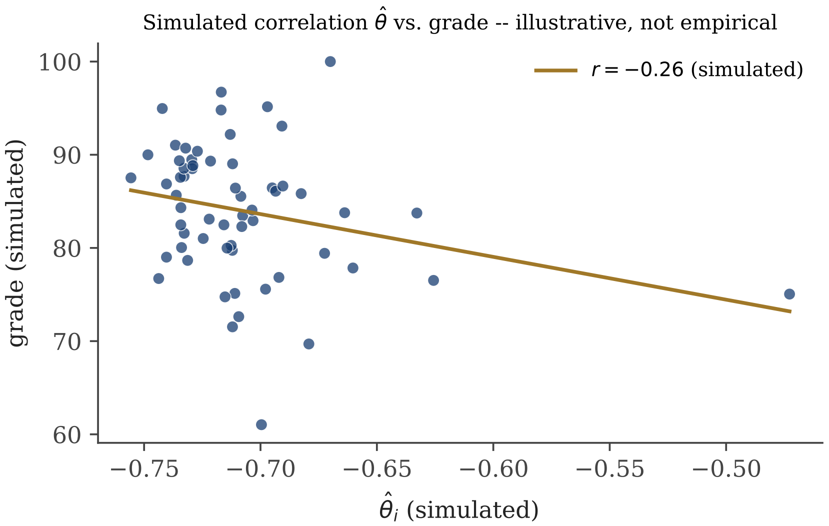

Figure 4. Simulated correlation between $\hat{\theta}$ and the grade, imposed by construction (c_i was defined as a decreasing function of the simulated emotional intensity e_i) to illustrate how this plot is read in the real interface.

3.1. Sensitivity analysis

The previous sections *claim* that $\hat{\theta}$ depends on sample size, on the wavelet family, and on the decomposition level, but up to this point did not show that numerically. This subsection does, running the same real code (`aplicar_dwt_posicional`, `calcular_indice_sentimiento`, `calcular_cramer_rao`) over controlled variations of the simulation above (`generar_figuras_sensibilidad_en.py`). Like the rest of this section, these are simulations, not course data.

Sensitivity to N. For $N \in \{8, 10, 15, 20, 30, 40, 60, 100, 150, 250\}$, with 20 independent replicates per value of N, we measure the bootstrap interval width $2\sqrt{\widehat{\mathrm{Var}}(\bar{\theta})}$ and the EEE (Figure 5). The right panel confirms the claim made in Section 4: the interval width decreases consistently and its variability across replicates is much larger for $N < 30$ than for large N. The left panel, on the other hand, reveals something Section 2.2.3 already anticipated: EEE stays between 0.96 and 1.00 across the *entire* range of N tested, with no clear improving trend — consistent with the fact that, under the Gaussian model, efficiency does not measure how good the pipeline is, but a property of the sample mean that holds almost always.

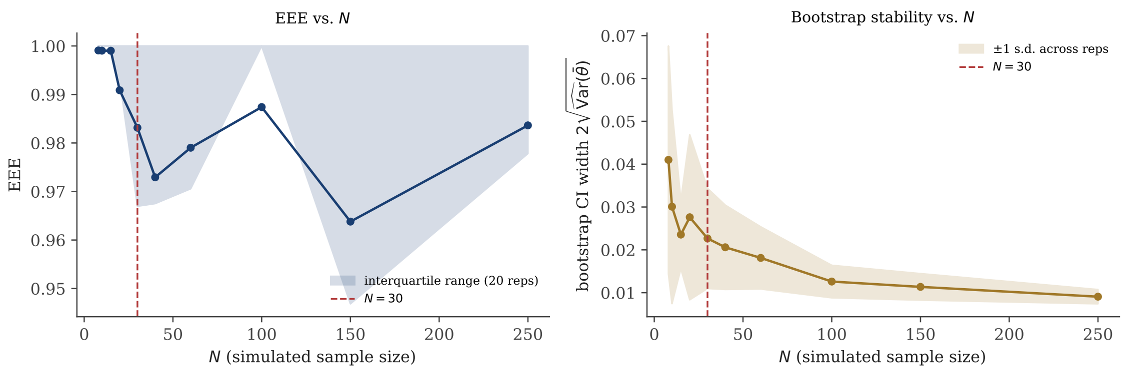

Figure 5. Sensitivity to N (20 replicates per value of N). Left: EEE stays high regardless of N. Right: the bootstrap interval width does decrease and stabilize with N, and is notably more dispersed across replicates for $N < 30$.

Sensitivity to wavelet family. On the same 60 simulated signals, we compute $\hat{\theta}$ with each of the six families offered by the real interface (`WAVELETS_DISPONIBLES` in `pages/sentimiento.py`): db4, db2, haar, sym4, coif2, and dmey (Figure 6). In aggregate distribution (median and spread), db4, sym4, coif2, and dmey look similar in the box plot, but the point-by-point Pearson correlation against db4 –how much two families agree on *which* student has a high or low $\hat{\theta}$, not just on the shape of the aggregate distribution– tells a more nuanced story: coif2 ($r = 0.80$) and dmey ($r = 0.72$) do agree closely with db4; sym4, despite its visually similar aggregate distribution, agrees more weakly at the individual level ($r = 0.59$, comparable to db2); db2 has a moderate correlation ($r = 0.77$) but with a tail of positive outliers; **haar is qualitatively different**: its median approaches zero and its interquartile range crosses the sign, with the lowest correlation in the group ($r = 0.50$). haar has a direct explanation: it is the family with the shortest filter (length 2), so its approximation/detail split captures a much finer, noisier notion of "local" than the longer-filter families. In practice, this means the default choice (db4) is not interchangeable with *all* of the menu's options: switching to haar can flip the sign of the reported index for the same text, and even a family with a similar aggregate distribution (sym4) can non-trivially reorder which students end up at the extremes.

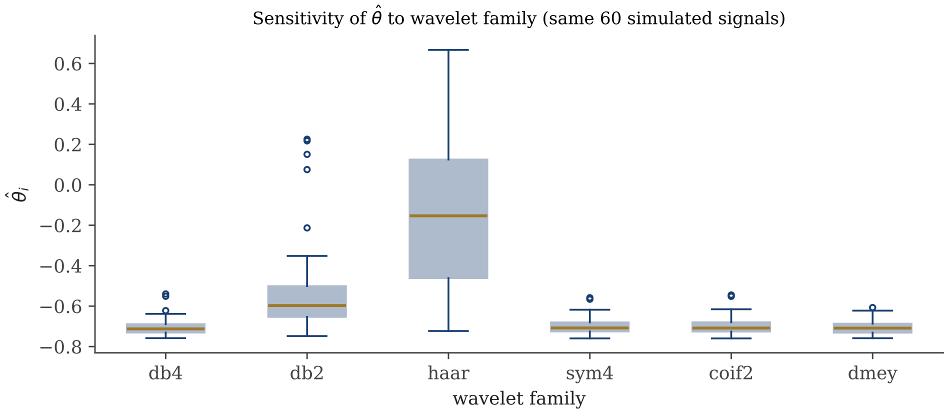

Figure 6. Distribution of $\hat{\theta}$ for the interface's six wavelet families, on the same 60 simulated signals. db4/sym4/coif2/dmey agree closely; haar –the shortest filter– is qualitatively different.

Sensitivity to level J: bounded by the shortest response. Varying $J \in \{1, \ldots, 5\}$ over this section's same short signals (12–54 tokens), the *effective* level used stayed fixed at 1 for all five requested values of J –`aplicar_dwt_posicional` bounds the level by the shortest response in the batch (Section 2.2.2)–, so $\hat{\theta}$ did not change at all as J varied (Figure 7, left panel). As a control, we repeated the experiment with long simulated responses (150–400 tokens): there the effective level does rise with J $(1, 2, 3, 4, 4)$ and $\hat{\theta}$ shifts noticeably, with growing dispersion at the higher levels (right panel).

Finding: the J parameter is effectively useless for typical surveys

With the response lengths simulated in this section (representative of short survey comments), the J parameter the user selects in the interface has no effect whatsoever: a single short response in the batch is enough for the effective level to stay bounded at 1 regardless of how high J is requested. The "Decomposition level J" slider in `pages/sentimiento.py` only matters, in practice, for surveys with consistently long responses. We added a warning in the interface (`pages/sentimiento.py`) that explicitly flags when this happens, instead of letting the user raise J with no visible effect and no explanation why.

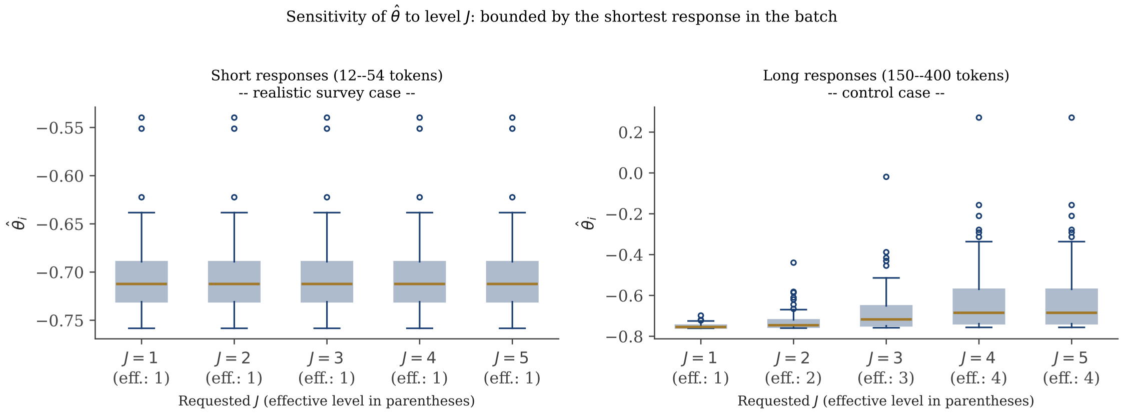

Figure 7. Sensitivity to J under two regimes: short responses (left, effective level stuck at 1) and long responses (right, effective level rises with J and $\hat{\theta}$ genuinely changes).

4. Impact

The system targets a generic teaching problem (not specific to one course or institution): the qualitative signal from an open-ended feedback survey is rarely analyzed with the same rigor as the quantitative signal from grades, despite being available in practically any course. By packaging token embeddings, DWT, and a Fisher/Cramér–Rao check into a reproducible pipeline with an interactive interface, the system lowers the barrier for an instructor without training in NLP or estimation theory to produce –and, above all, *correctly interpret*– a per-group aggregate sentiment index, including when *not* to trust it. That last part is, in practice, the least common component in similar tools: most educational-analytics dashboards report a number without stating what it mathematically guarantees and what it does not.

The honest limit on immediate reuse is the one already documented in the previous sections: the index $\hat{\theta}$ depends on the chosen embedding model, on the wavelet family and level, and on the survey's size –quantified in the sensitivity analysis of Section 3.1: the haar family is qualitatively different from the others, and J has no effect at all for short responses–; with $N < 30$ the bootstrap variance estimates are unstable, and the system itself warns about this in the interface. The textual interpretations generated by the `interpretacion.py` module are reading suggestions, not definitive conclusions, and should be checked against the instructor's judgment and, when warranted, institutional psycho-pedagogical support. As documented in Section 2.2.3, the Cramér–Rao check certifies the statistical coherence of $\bar{\theta}$ under the assumed model, not the quality of $\hat{\theta}$ as a measure of sentiment; that second question –the one that would actually matter for justifying institutional adoption– remains open and would require external comparison (human annotation, or another extraction pipeline) on real course data, which this article deliberately does not report. Until that validation exists, responsible reuse of the system is as an exploratory, supportive tool, not as the sole basis for a decision about a student.

Reuse

Code, sample data, 22 automated tests (`pytest`, run in CI on every *push*), and an MIT license are at github.com/Gabriel44svg/...-SIAMDA (see also Table 1). Architecture and dependency details are in Section 2.1.

5. Conclusions

This system integrates two signals from a course –grades and open-ended survey responses– under one dashboard, and adds to the latter a reproducible metric ($\hat{\theta}$, via token embeddings, positional projection, and DWT) together with a statistical check of the coherence of its aggregate mean (Fisher information and the Cramér–Rao bound, under Gaussian and Laplacian models). The result is an early-warning system that makes the link between a group's tone and its performance explicit rather than impressionistic, and that honestly documents what its statistical layer certifies and what it does not. Validating $\hat{\theta}$ against human annotation on real course data remains as future work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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